

Fatigue Strength Design Example 2

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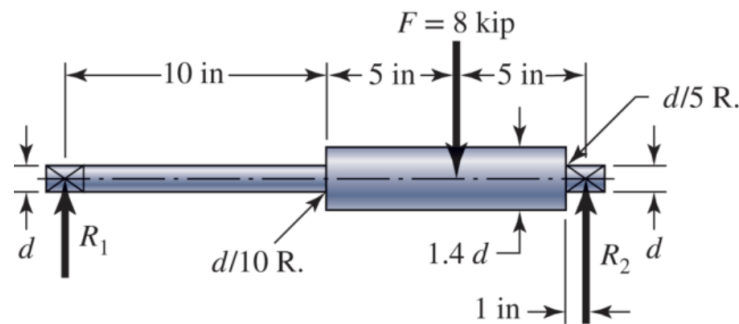
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1 Rotating Stepped-Shaft with Transverse Load

This problem is a compact walkthrough of fatigue design for a stepped, simply supported shaft in rotating bending. The external load fixes the support reactions and the bending-moment field; the stepped geometry then localizes stress at the shoulder where both the section modulus is smaller and a geometric notch exists. Our task is to connect these ingredients—equilibrium, bending, notch sensitivity, and S - N behavior—into a consistent estimate of alternating stress at the critical section and the corresponding safety against yielding and fatigue.

- 6-19** Bearing reactions R_1 and R_2 are exerted on the shaft shown in the figure, which rotates at 950 rev/min and supports an 8-kip bending force. Use a 1095 HR steel. Specify a diameter d using a design factor of $n_d = 1.6$ for a life of 10 hr. The surfaces are machined.



Problem 6-19

Figure 1: Equilibrium, bending diagram, and stress state identification at the shoulder. The problem is taken from Shigley.

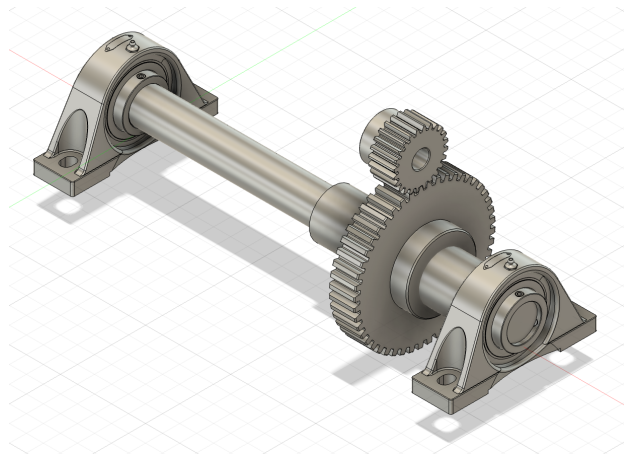


Figure 2: 3D View of a stepped shaft representing the 2D configuration shown above.

1.1 Step 1: Determine σ_a and σ_m

Critical-section analysis. To determine where the shaft is most highly stressed, we compare the bending stresses at two candidate sections: the large-diameter shoulder (A/B) and the small-diameter shoulder (C/D). Assuming the gear is located at a from the center of the left bearing and b from the center of the right bearing, $L = a + b$. We also parameterize the distance between the center of the left bearing and the gear center as c .

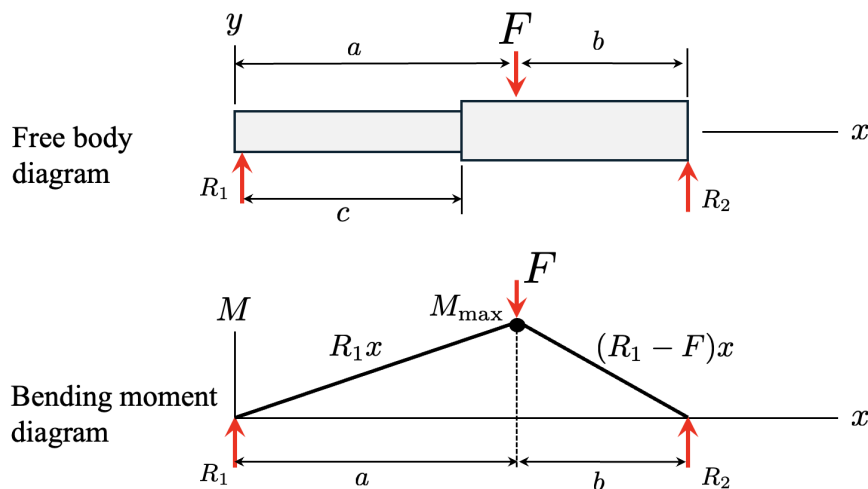


Figure 3: Free body diagram of the stepped shaft.

The bending moment diagram shows that the peak moment occurs under the applied load. Because surface bending stress scales with the local section modulus, any change of diameter along the span redistributes the stress severity. In stepped shafts this typically shifts the fatigue hotspot to the small-diameter shoulder, where the combination of higher nominal stress and a geometric notch (fillet) amplifies the local stress via K_t and K_f . Thus, the critical section needs to be identified after examining both the maximum bending moment location as well as the step locations.

First, recall that the bending stress at a section is

$$\sigma = \frac{32M}{\pi d^3}, \quad (1)$$

where M is the bending moment at that section and d is the local shaft diameter.

The reaction at the left support is

$$R_1 = \frac{Fb}{a+b}, \quad (2)$$

and the bending moment at a distance x from the left support is

$$M(x) = R_1 x. \quad (3)$$

At section A/B, located a distance a from the left support, the stress is

$$\sigma_{A/B} = \frac{32R_1 a}{\pi(1.4d)^3}, \quad (4)$$

since the diameter there is the larger $1.4d$.

At section C/D, located a distance c from the left support, the stress is

$$\sigma_{C/D} = \frac{32R_1c}{\pi d^3}. \quad (5)$$

Taking the ratio of these stresses gives

$$\frac{\sigma_{C/D}}{\sigma_{A/B}} = \left(1.4\right)^3 \cdot \frac{c}{a}. \quad (6)$$

From the shaft geometry, $c = 10$ in and $a = 15$ in. Substituting,

$$\frac{\sigma_{C/D}}{\sigma_{A/B}} = \frac{(1.4)^3}{1.5} \approx 1.83. \quad (7)$$

Since this ratio exceeds unity, we conclude that

$$\sigma_{C/D} > \sigma_{A/B}, \quad (8)$$

and therefore the critical location for fatigue design is the small shoulder at section C/D. As this section, the stress on a surface point experiences change in stress amplitude with the mean being zero.

At one orientation, the point may be in maximum tension; a half rotation later, the same point is in maximum compression. This periodic reversal generates a fully alternating stress cycle, as illustrated in Fig. 5.

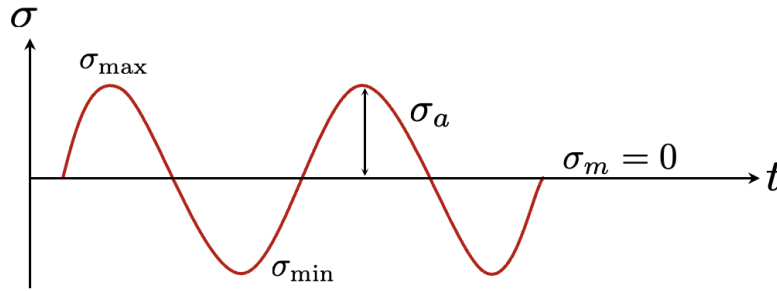


Figure 4: Fluctuation of surface stress in the rotating shaft.

Amplitude of alternating stress. For fluctuation in the normal stress,

$$\begin{aligned} \Delta\sigma_x &= K_{f,\text{axial}} \left| \frac{\sigma_{\max} - \sigma_{\min}}{2} \right|_{\text{axial}} + K_{f,\text{bending}} \left| \frac{\sigma_{\max} - \sigma_{\min}}{2} \right|_{\text{bending}} \\ &= 0 + 1 \left| \frac{\sigma_{\max} - (-\sigma_{\max})}{2} \right| = \sigma_{\max}. \end{aligned} \quad (9)$$

For the shear stress, there is no applied torque and no significant source of cyclic shear. Although the shaft experiences transverse shear due to the applied force, its magnitude is

relatively small compared to the bending stress at the outer surface. Thus, its effect can be neglected in fatigue analysis, and we obtain:

$$\begin{aligned}\Delta\tau_{xy} &= K_{f,\text{torsion}} \left| \frac{\tau_{\max} - \tau_{\min}}{2} \right|_{\text{torsion}} + K_{f,\text{shear}} \left| \frac{\tau_{\max} - \tau_{\min}}{2} \right|_{\text{shear}} \\ &= 0.\end{aligned}\tag{10}$$

The effective alternating stress amplitude is therefore given by

$$\sigma_a = \sqrt{\Delta\sigma_x^2 + 3\Delta\tau_{xy}^2} = \Delta\sigma_x = \sigma_{\max}.\tag{11}$$

Physically, σ_a represents the maximum stress intensity that the material experiences repeatedly during each revolution of the shaft. This is the critical quantity in fatigue analysis, since it defines the cyclic driving force that will govern crack initiation and eventual failure.

Mean stress. In rotating bending, the stresses at a surface point alternate symmetrically between tension and compression as the shaft completes a revolution. This symmetry means that the average (or mean) value of stress over a cycle is zero. In other words, for every tensile half-cycle there is an equal and opposite compressive half-cycle, resulting in no net mean stress.

$$\begin{aligned}\sigma_x &= K_{f,\text{axial}} \left(\frac{\sigma_{\max} + \sigma_{\min}}{2} \right)_{\text{axial}} + K_{f,\text{bending}} \left(\frac{\sigma_{\max} + \sigma_{\min}}{2} \right)_{\text{bending}} \\ &= 0 + 1 \left(\frac{\sigma_{\max} + (-\sigma_{\max})}{2} \right) = 0,\end{aligned}\tag{12}$$

$$\tau_{xy} = 0,\tag{13}$$

$$\sigma_m = \sqrt{\sigma_x^2 + 3\tau_{xy}^2} = 0.\tag{14}$$

Because the mean stress vanishes, mean-stress corrections such as Goodman or Gerber are unnecessary for the infinite-life check. Should any steady bias be introduced—e.g., a preload, misalignment, or residual stress—the analysis must be revisited with a mean-stress diagram.

The bending stress at the design shoulder section is

$$\sigma_{\max} = \frac{32M}{\pi d^3}, \quad M = \frac{Fbc}{L} \quad (\text{from reactions and the moment diagram}).$$

Including the fatigue stress–concentration factor in bending at the shoulder,

$$\sigma_a = K_f \frac{32Fbc}{\pi L d^3},$$

with K_f determined from notch sensitivity (below).

Two sensitivities deserve emphasis. First, the nominal amplitude varies with d^{-3} , so a few percent increase in d can produce a double-digit reduction in σ_a . Second, K_f depends

on material notch sensitivity through $\sqrt{a}$ and on the local fillet radius r . Since r scales with d here ($r = 0.1d$), any change in trial diameter also changes K_f , making iteration essential.

Using Appendix Figure A-15-9,

$$\frac{D}{d} = 1.4 \quad (15)$$

$$\frac{r}{d} = 0.1 \quad (16)$$

$$K_t = 1.65 \quad (17)$$

$$S_{ut} = 120 \text{ ksi} \quad (18)$$

$$S_y = 66 \text{ ksi} \quad (19)$$

$$\sqrt{r} = \sqrt{0.1d} \quad (20)$$

While we determined K_t with the given information on the ratios ($r/d = 0.1$ and $D/r = 1.4$), K_f can not be determined without specifying d . To proceed further, we **assume** $d = 2.0$ in, and find K_f from the following equation

$$\sqrt{a} = 0.246 - 3.08(10^{-3}) S_{ut} + 1.51(10^{-5}) S_{ut}^2 - 2.67(10^{-8}) S_{ut}^3 = 0.0477 \quad (21)$$

$$K_f = 1 + \frac{K_t - 1}{1 + \frac{\sqrt{a}}{\sqrt{r}}} = 1.59 \quad (22)$$

Adopting a trial diameter allows K_f to be determined consistently with the chosen d and $r = 0.1d$. This is standard practice for stepped shafts: assume d , compute K_f from the notch-sensitivity model, evaluate safety, and update d as required. Each update must propagate through all d -dependent terms—nominal stress ($\propto d^{-3}$), the Marin size factor $k_b(d)$, and $K_f(d)$ via $\sqrt{r}$ —to avoid optimistic results.

1.2 Step 2: Endurance limit using Marin factors

For 1095 HR steel (from Table A-20, use $S_{ut} \approx 120$ ksi, $S_y \approx 65$ ksi), unmodified endurance limit $S'_e = \frac{1}{2}S_{ut}$. Also, with machined surface and diameter in inches,

$$k_a = a S_{ut}^b = 2.0 S_{ut}^{-0.217} \quad (23)$$

$$k_b = 0.879 d^{-0.107} \text{ (or } 0.91 d^{-0.157}) \quad (24)$$

$$k_c = 1 \text{ (bending)} \quad (25)$$

$$k_d = 1 \quad (26)$$

$$k_e = 1 \quad (27)$$

With the assumed trial value of the design diameter $d = 2$ in,

$$S_e = k_a k_b k_c k_d k_e S'_e \approx 34.65 \text{ ksi.}$$

The nominal endurance limit S'_e represents the fatigue strength of a polished laboratory specimen, but real components rarely achieve this value. Surface finish, size effects, load

type, temperature, and reliability all act to reduce the fatigue resistance. These influences are captured by Marin factors, each of which scales down the baseline endurance strength. The mild power law in k_b means S_e changes slowly with d compared with the cubic change of σ_a . Unit consistency is critical here: the constants in k_a assume S_{ut} in ksi for the form used; mixing units can shift S_e by large factors.

1.3 Step 3: Factors of safety n_y and n_f

With $\sigma_m = 0$, fatigue safety factor at the trial diameter is

$$n_f = \frac{S_e}{\sigma_a} \approx 0.85.$$

A quick static check at the shoulder gives

$$n_y = \frac{S_y}{\sigma_a} = 1.63.$$

Safety must be checked against both static yielding and fatigue. The static check ensures that the shaft will not undergo plastic deformation even in the first cycle, while the fatigue check ensures long-term durability under cyclic loading. The contrast between $n_y > 1$ and $n_f < 1$ is common in fatigue design: a part may be strong enough to avoid plastic deformation but still too weak to resist cyclic damage. This is why fatigue design requires stricter criteria than static strength checks alone.

Note that 'design factor' should be sufficient to ensure the given finite life. As such as take the design factor as the ratio between 'allowable σ_a ' for N cycles, which we denote here as σ_{aN} using the given life, and the calculated σ_a using the given loading and geometric parameters. The expected design life is $N = 950(\text{rev. per minute}) \times 60 \text{ minute per hour} \times 10 \text{ hours} = 570000$ cycles.

1.4 Step 4: Finite-life evaluation and design factor

Use the two-point Basquin fit between ($N_1 = 10^3$, $\sigma_1 = fS_{ut}$) and ($N_2 = 10^6$, $\sigma_2 = S_e$):

$$a = \frac{(fS_{ut})^2}{S_e}, \quad (28)$$

$$b = -\frac{1}{3} \log_{10} \left(\frac{fS_{ut}}{S_e} \right) \quad (29)$$

$$N = \left(\frac{\sigma_a}{a} \right)^{1/b} \quad (30)$$

The finite-life allowable at the target life as

$$\sigma_{aN} = aN^b \approx 37.79 \text{ ksi.}$$

Define the *applied* design factor (Goodman/Gerber not needed since $\sigma_m = 0$) by

$$n_d = \frac{\sigma_{aN}}{\sigma_a}.$$

At the trial $d = 2$ in, this yields $n_d \approx 0.93 < 1.6$, so increase d and iterate, as outlined in the flowchart below.

Because $n_f < 1$, we cannot assume infinite life. Instead, we turn to the Basquin relation, which interpolates the S – N curve between the high-stress/short-life and low-stress/long-life asymptotes. The slope b captures the fatigue sensitivity to life, while a fixes the stress scale. The finite-life allowable $\sigma_{aN} = aN^b$ then sets the design factor $n_d = \sigma_{aN}/\sigma_a$. Because σ_a falls with d^{-3} while σ_{aN} changes only via $S_e(d)$ and $K_f(d)$, n_d increases rapidly with d , so only a modest diameter increase is usually needed to meet a finite-life target.

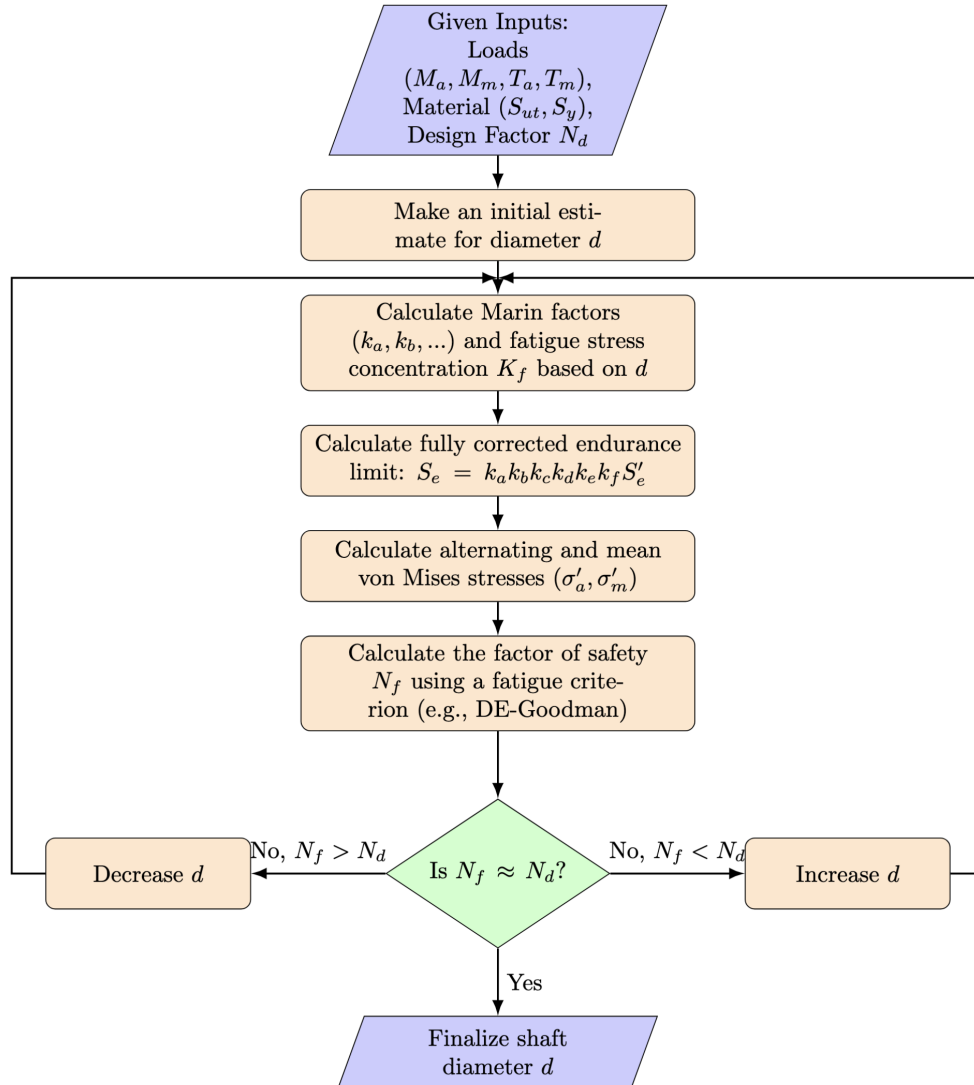


Figure 5: Flowchart showing the essential steps of the design process as an iterative process.

Result. Increasing the diameter to $d \approx 2.42$ in brings the design factor to $n_d \approx 1.6$ for the specified life.

1.5 Code

```
from numpy import *
import matplotlib.pyplot as plt

# Geometry and material information
S_ut = 120; S_y = 66;
L = 20; a = 15; b = 5; c = 10
d = 2.0;

# Step-1: Determine sigma_a and sigma_m
F = 8;
R1 = F*b/L;
M = R1*c;
sqrra = 0.246 - 3.08e-3*S_ut + 1.51e-5*(S_ut**2) - 2.67e-8*(S_ut**3)
r = 0.1*d;
Kt = 1.65;
Kf = 1 + (Kt - 1)/(1+sqrra/sqrt(r))

sigma_max = Kf*32*M/(pi*d**3)
sigma_min = -Kf*32*M/(pi*d**3)
sigma_a = (sigma_max - sigma_min)/2
sigma_m = (sigma_max + sigma_min)/2
print("M =",M)
print("Kt =",Kt, "sqrt(a) =", sqrra)
print("Kf =", Kf)
print("Sigma_a (from applied loading) = ", sigma_a)

# Step-2: Determine Se
ka = 2.0*S_ut**(-0.217)
kb = 0.91*d**(-0.157)
kc = 1; kd = 1; ke = 1
Se_prime = 0.5*S_ut
Se = ka * kb * kc * kd * ke * Se_prime
print("Se = ",Se)

# Step-3: Determine nf
if sigma_m <= 0:
    n_f = Se/sigma_a
elif sigma_m > 0:
    n_f = (sigma_a/Se + sigma_m/S_ut)**(-1)
print("FOS: n_f =", n_f)

# Step-4: Life-related calculation
N = 950*60*10;
print(N)
f = 1.06-2.8e-3*S_ut + 6.9e-6*S_ut**2
a = (f*S_ut)**2/Se
b = -(1/3)*log10(f*S_ut/Se)
sigma_aN = a*N**b;
print("a =", a, "b =", b, "f=",f)
print("Sigma_a (for expected N cycles) = ",sigma_aN)
print("FOS: n_d =",sigma_aN/sigma_a)
print("FOS: n_y =",S_y/sigma_a)
```